

S.No. 697953



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai.)

SATHYAMANGALAM - 638 401

ANSWER BOOKLET OF PERIODICAL TEST I / II

(B.E. / B.Tech. Programmes)

Reg. No.:

7 3 7 6 2 5 1 C S 1 1 4

Degree & Branch	CSE - B.E											
Course Code	2	2	M	A	1	0	1	Course Title	Engineering mathematics I			
Semester	I		  Name & Signature of the Invigilator with Date							For Office Use Only		
Date & Session	30-10-25 AN											
No. of Pages Used	8											

EVALUATION TABLE

SECTION A								SECTION B							
Q. No.	(i)	(ii)	(iii)	(iv)	(v)	(vi)	Total	Q. No.	(i)	(ii)	(iii)	(iv)	(v)	(vi)	Total
A1	1	0	3	2			6	B1	1	1	3	2			7
A2	1	0		2			3	B2							
A3								B3	0	0	1				1
SECTION A - TOTAL							9	SECTION B - TOTAL							8

SECTION C							
Q. No.	(i)	(ii)	(iii)	(iv)	(v)	(vi)	Total
C1	0						0
C2	1						1
C3	1						1
SECTION C - TOTAL							2

TOTAL MARKS (50)
(A+B+C)

19

Signature of the Examiner with date

M. K. Srinivas
4/11/25

Section - A

A1

i) d) 20

ii) d) 55

iii) $x: -4 \quad 1 \quad 2 \quad 3$
 $y: 4 \quad 6 \quad 8 \quad 10$

$$n=4$$

$$y = ax + b$$

$$a \sum x + nb = \sum y$$

$$a \sum x^2 + n \sum x = \sum xy$$

x	y	x^2	xy
-4	4	+16	-16
1	6	1	6
2	8	4	16
3	10	9	30
$\sum x = 2 \quad \sum y = 28 \quad \sum x^2 = 30 \quad \sum xy = 36$			

$$2a + 4b = 28 \rightarrow (1)$$

$$30a + 8 = 36 \rightarrow (2)$$

$$a = -0.769$$

$$b = 7.384$$

$$y = ax + b$$

$$y = (-0.769)x + 7.384$$

$$y = -0.769x + 7.384$$

$$y = 6.615x$$

iv)

$x: 0 \quad 1 \quad 2 \quad 3 \quad 4$
 $y: 1 \quad 1.8 \quad 3.3 \quad 4.5 \quad 6.3$

$$n=4$$

$$y = ax + b$$

$$a \sum x + nb = \sum y$$

$$a \sum x^2 + n \sum x = \sum xy$$

x	y	x^2	xy
0	1	0	0
1	1.8	1	1.8
2	3.3	4	6.6
3	4.5	9	13.5
4	6.3	16	25.2
$\sum x = 10 \quad \sum y = 16.9 \quad \sum x^2 = 30 \quad \sum xy = 47.1$			

$$10a + 4b = 16.9 \rightarrow (1)$$

$$30a + 40 = 47.1 \rightarrow (2)$$

$$a = -0.2 \quad 1.7414$$

$$b = -0.2 - 0.1285$$

$$y = 1.7414x + (-0.1285)$$

$$y = 1.7414x - 0.1285$$

$$y = 1.6129x$$

$$= 4 - 2 + 2$$

$$= 0 + (0 - 2) + 2(0 - 2)$$

$$= -2 + 2(-2)$$

$$= -2 - 4$$

$$S_2 = -6$$

A₂.

i) a) 6

ii) d) 36

iv)

$$A = \begin{bmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & -1 & -1 \\ -2 & 1 & -1 \\ -2 & 2 & 2 \end{bmatrix}$$

$$S_3 = \begin{vmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= 0 - 2(-2 + 2) - 2(1 + 1)$$

$$= -2(0) - 2(2)$$

$$= 0 - 4$$

$$S_3 = -4$$

$A \neq A^T$ it is non symmetric

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\lambda^3 - 3\lambda^2 - 6\lambda + 4 = 0$$

Eigen value is 3, 2, 6

Eigen vector

$$[A - \lambda I]x = 0$$

$$S_1 = 0 + 2 + 1 = 3$$

$$S_2 = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ -2 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ -2 & 1 \end{vmatrix}$$

$$= (2 + 2) + (0 - 2) + (0 - 2)$$

$$= 4 + (-2) + (-2)$$

$$\begin{bmatrix} 0 - \lambda & -2 & -2 \\ -1 & 1 - \lambda & 2 \\ -1 & -1 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\lambda = 2$$

$$\begin{bmatrix} -2 & -2 & -2 \\ -1 & 3 & 2 \\ -1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x_1}{\begin{vmatrix} 3 & 2 \\ -1 & 4 \end{vmatrix}} + \frac{x_2}{\begin{vmatrix} -2 & -2 \\ -1 & 4 \end{vmatrix}} + \frac{x_3}{\begin{vmatrix} -2 & -2 \\ -1 & 3 \end{vmatrix}}$$

$$\frac{x_1}{(12+2)} + \frac{x_2}{(-8-2)} + \frac{x_3}{(-6-2)}$$

$$\frac{x_1}{14} + \frac{x_2}{-10} + \frac{x_3}{-8}$$

$$x_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -3 & -2 & -2 \\ -1 & 4 & 2 \\ -1 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x_1}{\begin{vmatrix} 4 & 2 \\ -1 & 5 \end{vmatrix}} + \frac{x_2}{\begin{vmatrix} -3 & -2 \\ -1 & 5 \end{vmatrix}} + \frac{x_3}{\begin{vmatrix} -3 & -2 \\ -1 & 4 \end{vmatrix}}$$

$$\frac{x_1}{(20+2)} + \frac{x_2}{(-15-2)} + \frac{x_3}{(-12-2)}$$

$$\frac{x_1}{22} + \frac{x_2}{-17} + \frac{x_3}{-14}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 6$$

$$\begin{bmatrix} -6 & -2 & -2 \\ -1 & 7 & 2 \\ -1 & -1 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x_1}{\begin{vmatrix} 7 & 2 \\ -1 & 8 \end{vmatrix}} + \frac{x_2}{\begin{vmatrix} -6 & -2 \\ -1 & 8 \end{vmatrix}} + \frac{x_3}{\begin{vmatrix} -6 & -2 \\ -1 & 7 \end{vmatrix}}$$

$$x_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\textcircled{1} l = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$l = \sqrt{2}$$

$$\textcircled{2} l = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$l = \sqrt{2}$$

Normalized

$$N = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$D = N^T A N$$

$$N^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Section B

B 1)

i) a) 1955

ii) a) upward.

$$\text{iv) } p(x) = -2x^2 + 80x - 600$$

$$f(x) = -2x^2 + 80x - 600$$

$$x=0$$

$$p(x) = -600$$

$$(0, -600)$$

$$y=0$$

$$p(x) = -2x^2 + 80x - 600$$

$$a = -2, b = 80, c = -600$$

vertex

$$(h, k)$$

$$h = \frac{-b}{2a} = \frac{-80}{2(-2)}$$

$$h = \frac{80}{-2} = -40$$

$$h = -40$$

$$k = f(-40)$$

$$f(-40) = -2(-40)^2 + 80(-40) - 600$$

$$k = -7000$$

$$(h, k) = (-40, -7000)$$

$$\text{iii) } \begin{array}{ccccccc} x & -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ y & -5 & -1 & 0 & 1 & 3 & 4 & 4 \end{array}$$

B2)

$$\text{i) a) } \begin{bmatrix} 2 & -6 \\ 0 & 3 \end{bmatrix}$$

$$\text{ii) c) } 2$$

B3)

i) Positive definite

$$\text{ii) d) } (A^{-1})^{-1}$$

C1)

$$\text{i) b) } 3$$

~~ii)~~

C2)

$$\text{i) c) } a = 2.5, b = 10$$

C3) i) c) y increase more slowly than x

B1)

iii)

$$\begin{array}{ccccccc} x & -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ y & -5 & -1 & 0 & 1 & 3 & 4 & 4 \end{array}$$

$$n = 7$$

$$y = ax^2 + bx + c$$

$$\sum y = a \sum x^2 + b \sum x + nc$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

$$\sum x = 0, \sum x^2 = 28, \sum x^3 = 0$$

$$\sum x^4 = 196, \sum y = 6, \sum xy = 40$$

$$\sum x^2 y = 6$$

$$6 = 28a + 0b + 7c \rightarrow (1)$$

$$40 = 0a + 28b + 0c \rightarrow (2)$$

$$6 = 196a + 0b + 28c \rightarrow (3)$$

$$a = -0.2142$$

$$b = 1.4285$$

$$c = 1.7142$$

$$y = ax^2 + bx + c$$

$$y = -0.2142x^2 + 1.42857x + 1.7142$$

$$y = 2.9285$$

B3)

$$\text{iii)} \quad 6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$$

$$\begin{bmatrix} \text{coeff } x^2 & \frac{1}{2} \text{coeff } xy & \frac{1}{2} \text{coeff } xz \\ -\frac{1}{2} \text{coeff } xy & \text{coeff } y^2 & \frac{1}{2} \text{coeff } yz \\ \frac{1}{2} \text{coeff } xz & \frac{1}{2} \text{coeff } yz & \text{coeff } z^2 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -4 & 4 \\ -4 & 3 & -2 \\ 4 & -2 & 3 \end{bmatrix}$$